PHYSICAL SCIENCE
CHEMISTRY CYCLE TEST

Grade 10 **13 March 2026**

Time: 55 mins, no reading time 71 ~~66~~ **marks**

ASSESSOR: G. Main **MODERATORS: K. Bolton and T. Kirsten**

INSTRUCTIONS:

1. This question paper consists of 5 pages, and a data sheet. Please check your question paper is complete.
2. All questions are to be answered on the question paper.

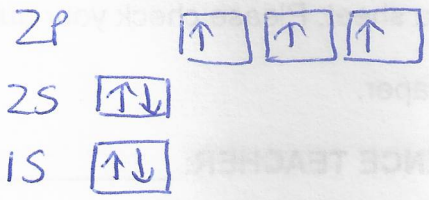
NAME: Memo **SCIENCE TEACHER:** _____

QUESTION 1

1.1 Complete the table below. (4)

Substance	Atomic number	Atomic Mass	Number of protons	Number of neutrons	Number of electrons
potassium	19	39 ✓	19 ✓	20	19
boron ion	5	11	5	6 ✓	2 ✓

1.2 Draw a labelled Aufbau diagram for the element **nitrogen**. (3)



✓ 7e⁻
✓ arrangement
✓ labels

1.3 Define *Pauli's exclusion principle*. (2)

Only two electrons can occupy an orbital and these must spin in opposite directions. ✓✓

1.4 Write down the electron configuration of the **chloride ion**. (3)

$(1s^2)(2s^2 2p^6)(3s^2 3p^6)$ ✓ ✓ ✓

1.5 An element **X** has the following electron configuration:

$1s^2 2s^2 2p^6 3s^2 3p^4$

For the element, **X**, write down the:

1.5.1 Group number on the periodic table. 16 ✓ (1)

1.5.2 Period number in the periodic table. 3 ✓ (1)

1.5.3 Number of valence electrons. 6 ✓ (1)

1.5.4 Valency. 2 ✓ (1)

1.5.5 Name of element **X**. Sulfur ✓ (1)

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QUESTION 2

2.1 Define *isotopes*. (2)

Atoms of the same element that have the same no. of protons but different no. of neutrons / mass number.

2.2 Study the unknown elements A to E:

A	B	C	D	E
$^{28}_{14}X$	$^{15}_{28}X$	$^{26}_{16}X$	$^{26}_{15}X$	$^{15}_{8}X$

2.2.1 Write down the letter(s) that represent(s) isotopes. Give a reason for your choice. (3)

B and D; same atomic number, different mass number

2.2.2 NAME the element represented by this isotope. phosphorus (1)

2.3 In nature, Cu has the following common isotopes.

Isotopes	Atomic mass	Abundance (%)
^{63}Cu	63	69
^{65}Cu	65	31

Calculate the relative atomic mass of Cu. (3)

RAM = (69% × 63) + (31% × 65) m

= 63,62 amu

[9]

QUESTION 3

3.1 Write FORMULAE for the following compounds:

3.1.1 hydrochloric acid HCl

3.1.2 gold (III) nitrite $Au(NO_2)_3$

3.1.3 magnesium permanganate $Mg(MnO_4)_2$

3.1.4 copper (I) sulfite Cu_2SO_3

3.1.5 lithium ethanoate $LiCH_3COO$ (10)

3.2 NAME the following compounds:

3.2.1 AgBr — Silver bromide

3.2.2 NaOH — Sodium hydroxide

3.2.3 Fe_2O_3 — iron (III) oxide

3.2.4 $(NH_4)_2CO_3$ — ammonium carbonate

3.2.5 $Pb(Cr_2O_7)_2$ — lead (IV) dichromate (12)

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QUESTION 4

$KC\ell O_3$ is used to generate oxygen in safety matches, fireworks and explosives.

$KC\ell O_3 \rightarrow KC\ell + O_2$

4.1 Give the name of $KC\ell O_3$. (2)

potassium chlorate

4.2 Balance the equation. (2)

2 **$KC\ell O_3$** → 2 **$KC\ell$** + 3 **O_2**

4.3 What is the oxidation number of **$C\ell$** in

a) $KC\ell$ — −1 (2)

b) $KC\ell O_3$ — +5 (2)

[8]

QUESTION 5

5.1 Balance the following equations:

5.1.1 $CaC\ell_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaC\ell$

5.1.2 $2\ FeC\ell_3 + H_2 \rightarrow 2\ FeC\ell_2 + 2HC\ell$

5.1.3 $2\ A\ell(OH)_3 + 3\ H_2SO_4 \rightarrow A\ell_2(SO_4)_3 + 6\ H_2O$ (5)

5.2 Consider the following word equations for an unbalanced chemical reactions:

Rewrite the equation using the correct chemical formulae and balancing numbers.

5.2.1 copper + nitric acid → copper (II) nitrate + nitrogen dioxide + water (5)

$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

✓ bal

5.2.2 ammonia + chlorine → nitrogen trichloride + hydrochloric acid (5)

$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$

✓ bal

~~[10]~~ [15]